DP Graded Quiz M2

Question 1

Compute the variance of $\int_0^3 t dW_t$.

- 1
- 27
- 9
- 3

Question 2

(**Note:** Use Python to solve this question.) Compute the price of an American call with $u = 1.5$, $d = 1/u, T = 5 = N, r = 0$, $K = 31$, and $S_0 = 36$.

- 12.032
- 8.6917
- 14.7315
- 16.3529

Question 3

Compute the payoff at maturity of an Asian put option (the payoff is defined by the maximum between the strike price minus the average of the underlying values and zero) if the path is {d,d}. Data: 2-step binomial tree with $u = 1.2$, $d = 0.8$, $K = 50$, and $S_0 = 50$.

- 9.33
- 0
- 16

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- 4

Question 4

(**Note:** Use Python to solve this question.) Compute the Δ_2^{dd} of an American call with $u = 1.2$, $d = 1/u$, $T = 5 = N$, $r = 0$, $K = 45$, and $S_0 = 45$.

- 0
- 0.5847
- 0.1622
- 0.5679

Question 5

If

$$dX_t = 2dt + 5dW_t$$

and $Y_t = tX_t$, then $d[Y, Y]_t$ is equal to:

- $5tdW_t$
- $5dt$
- dt
- $25t^2dt$

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Question 6

(**Note:** Use Python to solve this question.) Compute the price of an American call with $u = 1.2$, $d = 1/u, T = 5 = N, r = 0$, $K = 105$, and $S_0 = 45$.

- 0.1353
- 0
- 0.5029
- 10.489

Question 7

The mean of $\int_0^2 t^2 dW_t$ is:

- 0
- 1
- 2
- 5

Question 8

(**Note:** Use Python to solve this question.) Compute the Δ_5^{ddddd} of an American call with $u = 1.2$, $d = 1/u, T = 5 = N, r = 0$, $K = 45$, and $S_0 = 45$.

- 0.2975
- 0
- 0.5454
- 0.5847

Question 9

If

$$dX_t = 2dt + 6dW_t$$

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and $dY_t = 4dt + dW_t$, then $d[X, Y]_t$ is:

- $6dt$
- $-4dt$
- $4dt$
- $5dt$

Question 10

(**Note:** Use Python to solve this question.) Compute the price of an American put with $u = 1.5$, $d = 1/u$, $T = 5 = N$, $r = 0$, $K = 31$, and $S_0 = 36$.

- 8.9209
- 9.9203
- 9.3419
- 9.7315

Question 11

The distribution of $\int_0^2 t^3 dW_t$ is:

- $N(1, 1)$
- $N(0, 1)$
- $N(1, 4)$
- $N(0, 128/7)$

Question 12

Delta measures how....

- the option price changes with the underlying price.
- the option price changes with the risk-free rate.

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QUESTIONS

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

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- the option price changes with the option maturity.
- the option price changes with the underlying volatility.

Question 13

If

$$dX_t = 2dt + 5dW_t$$

and $Y_t = X_t^2$, then dY_t is equal to:

- $(4X_t)\,dt + 10X_t\,dW_t$
- $(25)\,dt + 10X_t\,dW_t$
- $(2X_t + 25)\,dt + 10X_t\,dW_t$
- $(4X_t + 25)\,dt + 10X_t\,dW_t$

Question 14

Consider a Wiener process, W_t with volatility equal to 10. Then:

- $\frac{W_t}{10}$ is a standard Wiener process.
- $\frac{W_t}{\sqrt{(10)}}$ is a standard Wiener process.
- $\frac{W_t}{1000}$ is a standard Wiener process.
- $\frac{W_t}{100}$ is a standard Wiener process.

Question 15

In general, the accuracy of Monte Carlo methods...

- increases with the number of simulations only for Asian options.

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- decreases with the number of simulations.
- is not dependent on the number of simulations.
- increases with the number of simulations.

Question 16

(**Note:** Use Python to solve this question.) Compute the price of an American put with $u = 1.2$, $d = 1/u$, $T = 50 = N$, $r = 0$, $K = 45$, and $S_0 = 45$.

- 20.12
- 32.14
- 21.53
- 23.28

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